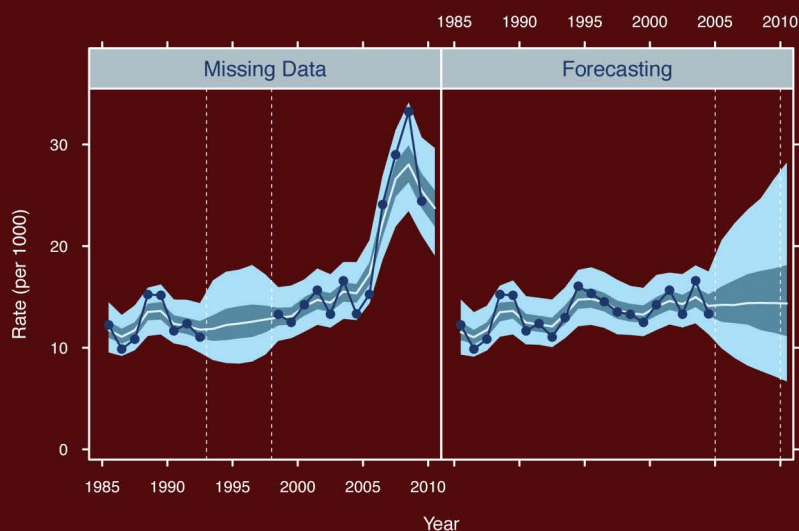


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Bayesian Demographic Estimation and Forecasting



John Bryant
Junni L. Zhang



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By
John Bryant
Junni L. Zhang



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Preface

This book presents new methods for answering questions such as

- How long can a newborn baby expect to live, given the ethnicity and income of the baby's parents?
- What proportion of future increases in health spending will be due to population aging?
- How should alternative estimates of population size be reconciled when these estimates do not agree?

The methods combine ideas from demography with ideas from statistics—especially a subfield known as Bayesian statistics.

The methods presented in this book form a coherent whole. We start with a simple framework, in which we model a single demographic series such as births or deaths. We then progressively expand the framework, to the point where we can deal with an entire system of interacting demographic series, measured through multiple, unreliable data sources.

Statistical demography is an interdisciplinary effort, involving statisticians and demographers, but also economists, epidemiologists, sociologists, and many others. We have tried to make the book useful and accessible to a broad audience. We assume only minimal knowledge of statistics, and no previous knowledge of demography. We focus on the underlying ideas, rather than the mathematical details.

We have developed a set of *R* packages implementing the methods in the book. The website for the book, www.bdef-book.com, has the packages along with data and code for all the examples. Most of the models could, alternatively, be fitted using general-purpose Bayesian computing environment such as BUGS or Stan. When fitting the models in this book, our packages are probably easier to work with, since they were designed specifically for the types of data and models used in the book. BUGS and Stan, however, offer more possibilities for extending the models.

The book is not a manual for our packages, and does not itself include any *R* code. Our hope is that readers wishing to find out about the new methods will use the book to learn about the principles, and will use the materials on the website to learn about the packages.

We focus on our own particular approach to statistical demography, and make no attempt to review the field as a whole. Focusing on just one approach

leads to a shorter and more unified book. However, statistical demography is a diverse, fascinating, and rapidly expanding field, and we point readers towards other approaches by giving suggested readings at the end of each chapter.

Acknowledgment

Jenny Harlow wrote the *C* code in our packages, and advised on software design. Jakub Bijak, Patrick Graham, Kirsten Nissen, Charlotte Taglioni, and Feifei Wang have worked with us on various aspects of Bayesian demography. The anonymous reviewers of the book gave numerous valuable suggestions. John Kimmel of CRC Press has been an ever-helpful guide during the writing of the book, and Robin Lloyd-Starkes has supervised the production process.

Statistics New Zealand has supported John Bryant in developing the software, and provided time and flexibility for working on the book. (The views expressed in this book are, however, entirely our own, and should not be attributed to Statistics New Zealand or to any other organization.) The Royal Society of New Zealand and the Center for Statistical Science and Guanghua School of Management at Peking University have funded several visits to Peking University by John Bryant, and the Guanghua School of Management has also funded a visit to New Zealand by Junni Zhang. We began our collaboration on Bayesian demography at the Small Area Estimation Conference 2013, and have received advice and encouragement from the small area estimation community since that time.

We thank our families for the love and support they have shown us during the writing of the book, and we dedicate the book to them.

Introduction

Goal 17.18 ... to increase significantly the availability of high-quality, timely and reliable data disaggregated by income, gender, age, race, ethnicity, migratory status, disability, geographic location and other characteristics relevant in national contexts.

— United Nations General Assembly

In September 2015, the United Nations General Assembly adopted the 2030 Agenda for Sustainable Development. The 2030 Agenda refers repeatedly to the need to compare progress made by smaller populations within the national population. Goal 1.1, for instance, emphasizes the importance of comparing poverty rates by sex, age, employment status, and geographical location (urban/rural). Goal 2.3 states that the productivity of small-scale food producers should be compared across sex, ethnicity, and occupation. Goal 17.18 calls more generally for the production of disaggregated statistics about development.

The focus on detailed comparisons in the 2030 Agenda is typical of contemporary analysis and policy making. Demand for disaggregated statistics is growing, not just among international organizations, but also among national governments, researchers, businesses, local authorities, and members of the public. Proponents of evidence-based policies want to target policies to tightly-defined groups. Strategists want to divide markets into ever-finer segments. Individuals want statistics about their own local area, age group, or ethnicity.

The demand for disaggregation extends to demographic forecasts. Users of demographic forecasts are no longer satisfied with national-level numbers. They want forecasts to be disaggregated by all the same variables that are used for estimates.

Producing reliable disaggregated demographic estimates and forecasts is difficult. Recent developments in statistical methodology have, however, opened up new possibilities. This book presents a new approach that combines ideas from a branch of statistics known as Bayesian statistics with ideas from mathematical demography. In this chapter, we describe some distinctive features of this new approach. But first, we illustrate our approach with an example.

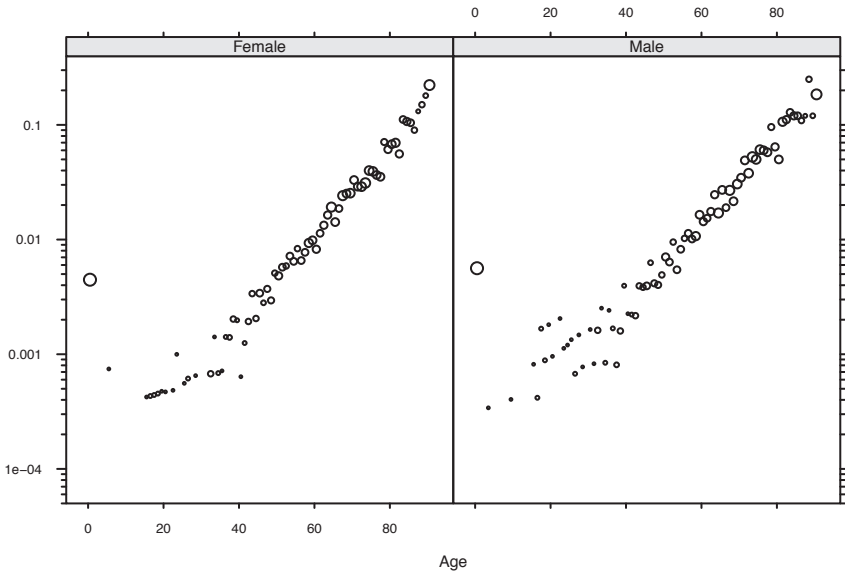


FIGURE 1.1: Estimates of Māori mortality rates in 2013 by single year of age. The graphs are drawn on a log scale. Each bubble shows the number of deaths divided by the corresponding population at risk. The size of each bubble is proportional to the number of deaths at that age. To preserve confidentiality, the deaths data have been randomly rounded to multiples of 3.

1.1 Example: Mortality Rates for Māori

Māori are the indigenous people of New Zealand. Measuring Māori progress towards lower mortality rates is a fundamental part of measuring New Zealand’s overall social and economic performance.

Figure 1.1 shows estimates of the mortality rates for Māori in 2013, constructed by dividing the number of deaths in each age-sex group by the number of people in that group. In 2013, the Māori population of New Zealand was around 0.7 million, and there were just over 3,100 Māori deaths. The age groups in Figure 1.1 are single years, that is, age 0, age 1, age 2, and so on. The estimates are shown on a log scale. The diameters of the bubbles are proportional to the number of deaths: for instance, the bubble for 0-year-old males, for whom there were approximately 160 deaths is about 10 times larger than the bubble for 1-year-olds, for whom there were approximately 15 deaths.

New Zealand’s official statistics agency, Stats NZ, has randomly rounded the deaths data to multiples of base 3, to protect confidentiality. Because the numbers of deaths in some cells are small, this rounding has a noticeable

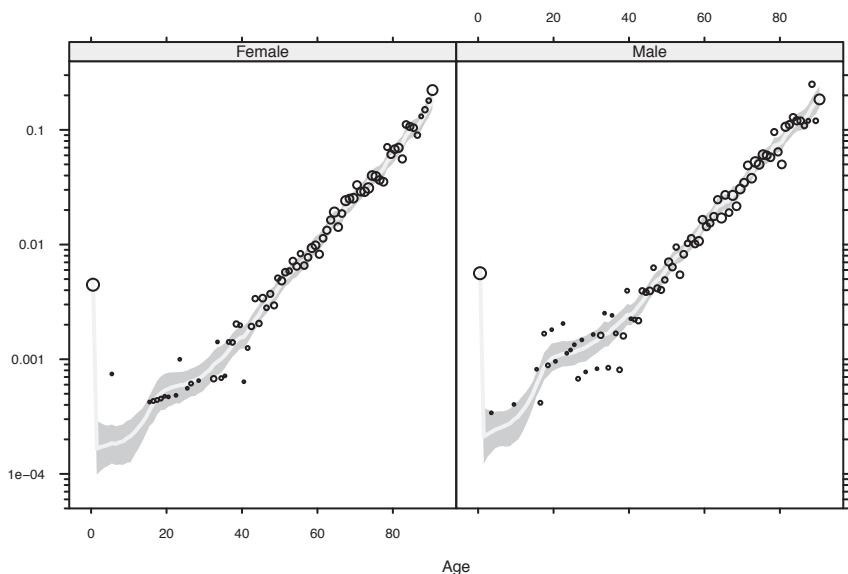


FIGURE 1.2: Model-based estimates of Māori mortality rates in 2012-2014 by single year of age. The graphs are drawn on a log scale. The gray bands are 95% credible intervals, and the pale lines are point estimates, from the model. The bubbles are simple estimates from [Figure 1.1](#).

impact on the data. The apparent layering of mortality rates below age 40, particularly for males, is an artefact of the random rounding.

We would like to measure the underlying risk of dying faced by Māori in each age-sex group in 2013. The simple estimates shown in [Figure 1.1](#) do not get these risks quite right. One reason is that random rounding distorts the rates. But the main reason is that observed number of deaths is affected by random variation, particularly when the numbers are small, which makes it a noisy indicator of underlying risks. In some years, for instance, some young age groups experience no deaths at all. We would not say that, in those years, the actual risk of dying was zero.

The models in this book provide ways of extracting signal from noisy data, so that we can see the underlying risks more clearly. [Figure 1.2](#) shows results from one such model. We in fact fitted the model to data for 2006 as well as 2013 to boost sample sizes, but we show results only for 2013 here.

The gray bands in [Figure 1.2](#) are 95% credible intervals. If the assumptions underlying the model are approximately correct, then there is a 95% chance that the true underlying rates lie within the credible intervals. Wider bands imply greater uncertainty. The pale lines in the middle of the credible intervals are point estimates, that is, single-number summaries of the results.

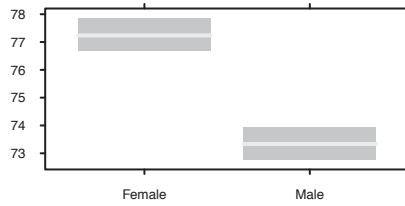


FIGURE 1.3: Māori life expectancy in 2013. The gray bands are 95% credible intervals, and the pale lines are point estimates.

The modeled estimates more or less follow a straight line from about age 30. Between the late teenage years and late 20s, the modeled rates are higher than we would expect from the straight line, particularly for males. This is the ‘accident hump’, a feature of most contemporary mortality profiles. The credible intervals are much wider for the young than for the old, reflecting the fact that there are fewer deaths, and therefore less data for estimating rates, for the young than the old.

Figure 1.3 shows estimates of life expectancy derived from the estimates of mortality rates. The uncertainty about the rates propagates through to the life expectancies. Here, as in the previous figures, the gray bands show 95% credible intervals.

The model used to generate the estimates in Figures 1.2 and 1.3 in fact does more than just estimate mortality rates. It also estimates the original, unrounded death counts, to remove any biases due to rounding. These estimates also come with measures of uncertainty. Table 1.1, for instance, shows estimates of the true number of deaths of 20 year old males. Once again, the answers are expressed in terms of probabilities. The table reports, for instance, that there is a 0.38 probability that the true, unconfidentialized number of deaths of 20 year olds was 6, and a 0.10 probability that it was 8.

TABLE 1.1

Estimated number of 20 year old males in 2013.

Value	4	5	6	7	8
Probability	0.08	0.22	0.38	0.22	0.10

1.2 Our Approach to Demographic Estimation and Forecasting

The methods in this book, like traditional demographic methods, focus on groups of people, as defined by characteristics such as age, sex, and region, rather than on individuals. Focusing on groups rather than individuals has disadvantages. For example, it limits the number of variables we can consider. But it also has advantages. It is well-suited to understanding change over time, and doing forecasting. It allows us to exploit datasets, such as confidentialized tables or historical time series, that have information on groups rather than individuals. And it matches the needs of many of the ultimate users of analyses. People planning schools and hospitals, or analyzing trends in mortality and population size, are interested in groups, not individuals.

We draw heavily on traditional demographic techniques for describing populations, from Lexis triangles (defined in [Section 3.3](#)), to origin-destination formats for migration ([Section 4.3](#)), to demographic accounts ([Chapter 5](#)). Compared with most traditional demography, however, the methods in this book are more formal and mathematical. We try to make our assumptions more explicit than demographers have traditionally done.

Taking a more formal approach requires making a clear distinction between (i) the true quantities that we wish to infer, and (ii) the data on which our inferences are based. For instance, we need to distinguish clearly between the true population counts that we are ultimately interested in, and the imperfect census data that are available to us for estimation.

Taking a formal approach also requires us to distinguish between (i) the underlying risks or propensities, and (ii) the random events governed by these risks or propensities. For instance, we need to distinguish between the probability that a woman aged 30 will give birth over the following year, and the actual proportion of women aged 30 who give birth.

Our approach to constructing formal statistical models draws on a branch of statistics known as Bayesian statistics. We say more about Bayesian statistics in [Chapters 8](#) and [9](#), but it is an alternative to ‘classical’ statistics that has become increasingly influential across a range of disciplines. Bayesian statisticians are willing to use probabilities to characterize all forms of uncertainty, including uncertainty due to randomness, and uncertainty due to limited knowledge. Bayesian methods deal well with complex models, which makes them well suited to the problems of modern statistical demography.

The particular mix of mathematical demography and Bayesian statistics that we develop in this book emphasizes some features in particular:

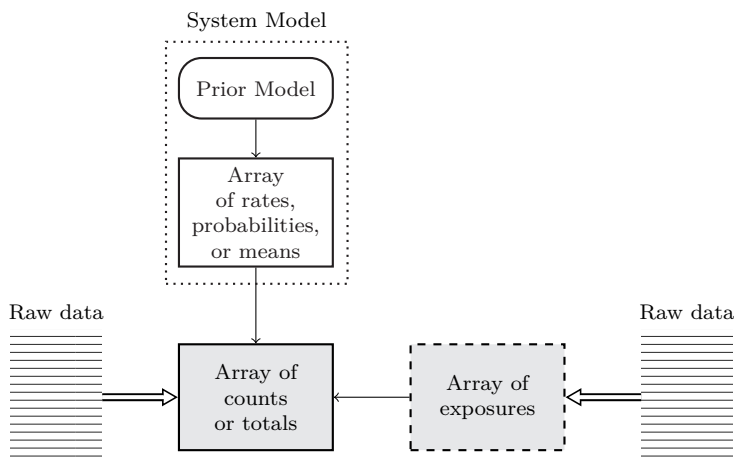


FIGURE 1.4: Inferring rates, probabilities, or means from a reliable dataset. The two gray rectangles are ‘demographic arrays’, constructed from raw individual-level data. The gray rectangles are partly or fully observed, and are treated as if they were completely error-free. We use the array of counts or totals to infer the array of rates, probabilities, or means, with the help of the prior model. The array of exposures is optional.

1. Disaggregation
2. Forecasting
3. Unreliable data
4. Demographic systems

As we will discuss in [Section 5.1](#), a demographic system is a set of interrelated demographic series, such as population, births, deaths, and migration. Some demographic applications involve all four features—disaggregation, forecasting, unreliable data, *and* demographic systems—but others do not. We present three general frameworks, aimed at progressively more complicated applications. The first framework, which is the simplest, is summarized in [Figure 1.4](#).

The framework is used when estimating underlying rates, probabilities, or means from data on counts or totals. The data and the underlying rates, probabilities, or means are organized in the form of demographic arrays. ‘Demographic array’ is our term for counts, rates, or other values cross-tabulated by dimensions such as age, sex, and time.

We assume that the data to hand are sufficiently reliable that they can be treated as if they were completely error-free. In other words, we pretend that the reported values for births, deaths, population, or any other demographic quantity are identical to the true values. This assumption sounds extreme when stated in this way, but it is in fact standard in much statistical modeling.

It is a sensible strategy when the errors are small, and have only a minor effect on results.

The framework of [Figure 1.4](#) does, however, allow for the possibility of gaps in the data. Within the framework, the raw data, and hence the arrays of counts or totals that are assembled from this data, may not have values for every combination of age, sex, time, or other dimensions that we are interested in. When values are missing, we infer them as part of the overall estimation process.

Because the framework allows for missing data, it allows for demographic forecasts. In a forecast, the data has the format

Observed	Not observed
----------	--------------

time →

We have data up to a certain time, and then no data past that time. Consider, for instance, the problem of forecasting birth rates for 2019–2028, based on data for 1990–2018. One way of formulating the problem is that we are estimating birth rates for the combined period 1990–2028, but have missing data for 2019–2028.

When estimating underlying rates, probabilities, or means it is usually wise to take into account regularities in the data, such as a shared tendency for death rates to rise with age. Similarly, we may be able to improve our estimates by incorporating information from other settings, such as information on plausible ranges. Strengthening estimates through the sharing of information is the job of the prior model. In the framework of [Figure 1.4](#), the prior model and the model describing how counts or totals are generated given rates, probabilities, or means are referred to jointly as the ‘system model’.

When modeling events, we generally need to take account of the size of the population that is at risk of experiencing the event. When modeling deaths by region, for instance, we would usually want to include a measure of regional population size. We refer to measures of population at risk as ‘exposures’. In the framework of [Figure 1.4](#), we treat exposures as error-free.

Although the framework of [Figure 1.4](#) has wide application, sometimes the measurement errors in the data are *not* small, and ignoring them *would* materially affect the conclusions from the analysis. In such cases, we need to shift to the framework of [Figure 1.5](#).

The key difference between the frameworks of [Figures 1.4](#) and [1.5](#) is that in [Figure 1.5](#) the true array of counts or totals is no longer treated as directly observed. Instead, all that is observed is one or more arrays of unreliable data. We must infer true counts or totals from the unreliable data.

To do so, we have to make some assumptions about the nature of the unreliability of the data. These assumptions are captured by ‘data models’. Each dataset has one data model. The data model expresses how the dataset is generated given the true counts or totals. A simple data model, for instance,

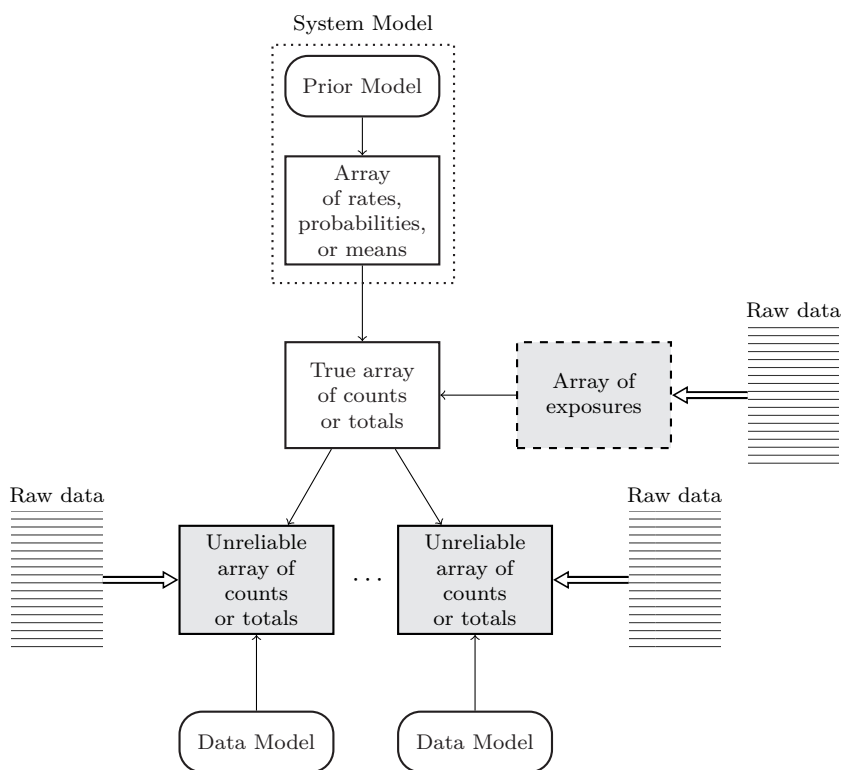


FIGURE 1.5: Inferring rates, probabilities, or means, plus the associated true counts or totals. In contrast to the framework of [Figure 1.4](#), the array of counts or totals has to be inferred from one or more unreliable datasets, which requires specifying models for the relationship between the true counts or totals and the data.

might say that a dataset captures 50% of the true events, so that if 20 births occur during a period, approximately 10 of these will appear in the dataset.

The system model is completely unaffected by the change in the status of the array of counts or totals. In the framework of [Figure 1.5](#), just like that of [Figure 1.4](#), the system model is composed of the model describing how true counts or totals are generated given rates, probabilities, or means, plus a prior model capturing expected trends and patterns.

The exposure term in the framework of [Figure 1.5](#) is also identical to that of [Figure 1.4](#). It is optional, but if included, is treated as error-free.

The frameworks of [Figures 1.4](#) and [1.5](#) both deal with single demographic series. With the framework of [Figure 1.6](#), we move to looking at an entire demographic system.

Instead of a single array of counts or totals, we have multiple arrays of counts. Each array represents a demographic series. There is always an array containing population counts. The remaining arrays contain counts of events such as births, deaths, and migrations that specify how people enter, exit, or move around the system. These counts are treated as unobserved, and must be inferred as part of the estimation process.

The arrays of counts together form a demographic account. The elements within a demographic account conform to a fundamental accounting identity: population at the end of each period equals population at the beginning plus entries minus exits.

The system model for each series in the demographic account is essentially the same as the system models in the two previous frameworks. The main difference is that exposure is not treated as fixed and known. Instead, it is derived from the array of population counts during estimation.

Each array of counts can be described by one or more datasets, and each dataset has its own data model. An array can even have no datasets, in which case it is inferred from the other arrays.

1.3 Outline of the Rest of the Book

[Part I](#) covers the demographic aspects of our approach: demographic arrays, demographic accounts, and their relationship with individual-level biographies. [Part II](#) covers the Bayesian statistical methods that we use to infer the unknown quantities in [Figures 1.4–1.6](#).

[Part III](#) looks at models falling within the framework of [Figure 1.4](#), where we have a single array of counts or totals that we treat as error free. [Part III](#) starts with a short chapter where we describe the framework in more detail. There are then three applications chapters, each presenting a case study based on the framework. [Parts IV](#) and [V](#) deal with the frameworks of [Figures 1.5](#) and [1.6](#). They follow the same introduction-applications format as [Part III](#).

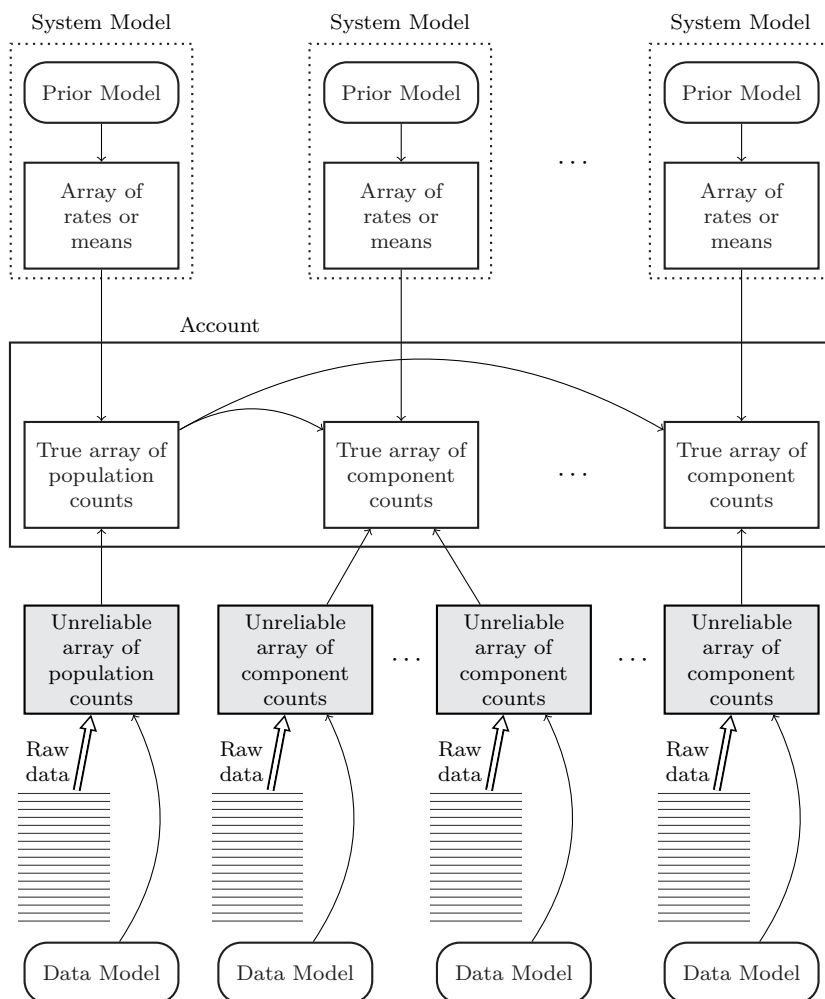


FIGURE 1.6: Inferring a demographic account. The gray rectangles are partly or fully observed; everything else is unknown and needs to be inferred. Each series in the demographic account can be measured by one or more datasets, or may not be measured by any dataset. Each dataset has its own data model. Each demographic series has its own system model.

We conclude the book with some reflections on the future development of the methods.

1.4 References and Further Reading

The opening quote and description of the goals of the 2030 Agenda for Sustainable Development come from United Nations General Assembly (2015).

The data on Māori mortality rates come from the table *Deaths by age and sex (Maori population) (Annual-Dec)* from the *Infoshare* database on the Stats NZ website. The data on Māori population come from the table *Estimated resident population (ERP), national population by ethnic group, age, and sex, 30 June 1996, 2001, 2006, and 2013* from the *NZ.Stat* database on the Stats NZ website. The data were downloaded on October 24, 2016. The code and data for the model are available at the website for this book, www.bdef-book.com.

Figures 1.1–1.3 were drawn using functions from *R* package **lattice** (R Core Team, 2016; Sarkar, 2008). Figures 1.4–1.6 were drawn using the **TikZ** package (Tantau, 2008). We use **lattice** and **TikZ** extensively through the rest of the book.

Introductions to demographic methods include Preston et al. (2001) and Wachter (2014). Demographic texts with a greater focus on estimation include Smith et al. (2013) and Swanson and Tayman (2012). Alho and Spencer (2005) is a wide-ranging textbook on statistical demography. Bijak and Bryant (2016) review Bayesian demography.

In 2015, the United Nations Population Division used Bayesian methods for its global population projections (Gerland et al., 2014; Raftery et al., 2014; UN Population Division, 2015). Abel et al. (2016) critique the projections.

Gelman and Hill (2007); Lynch (2007) are textbooks on Bayesian statistics aimed at social scientists, and Gelman et al. (2014) and McElreath (2016) are texts on Bayesian statistics in general.

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